

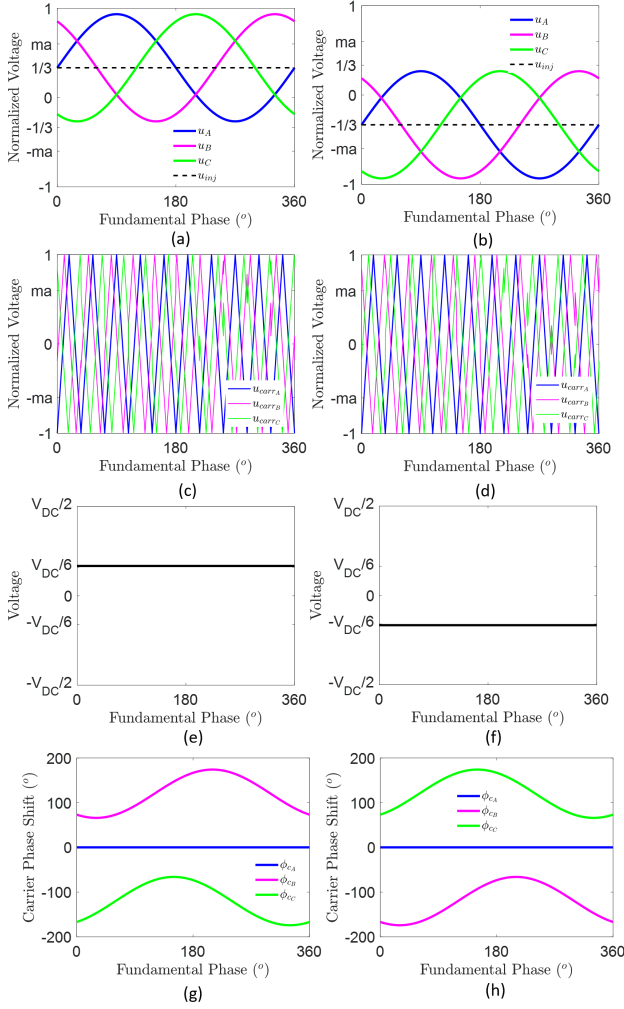
Mitigation of Common-Mode Voltage in 2-Level 3-Phase Voltage Source Inverters

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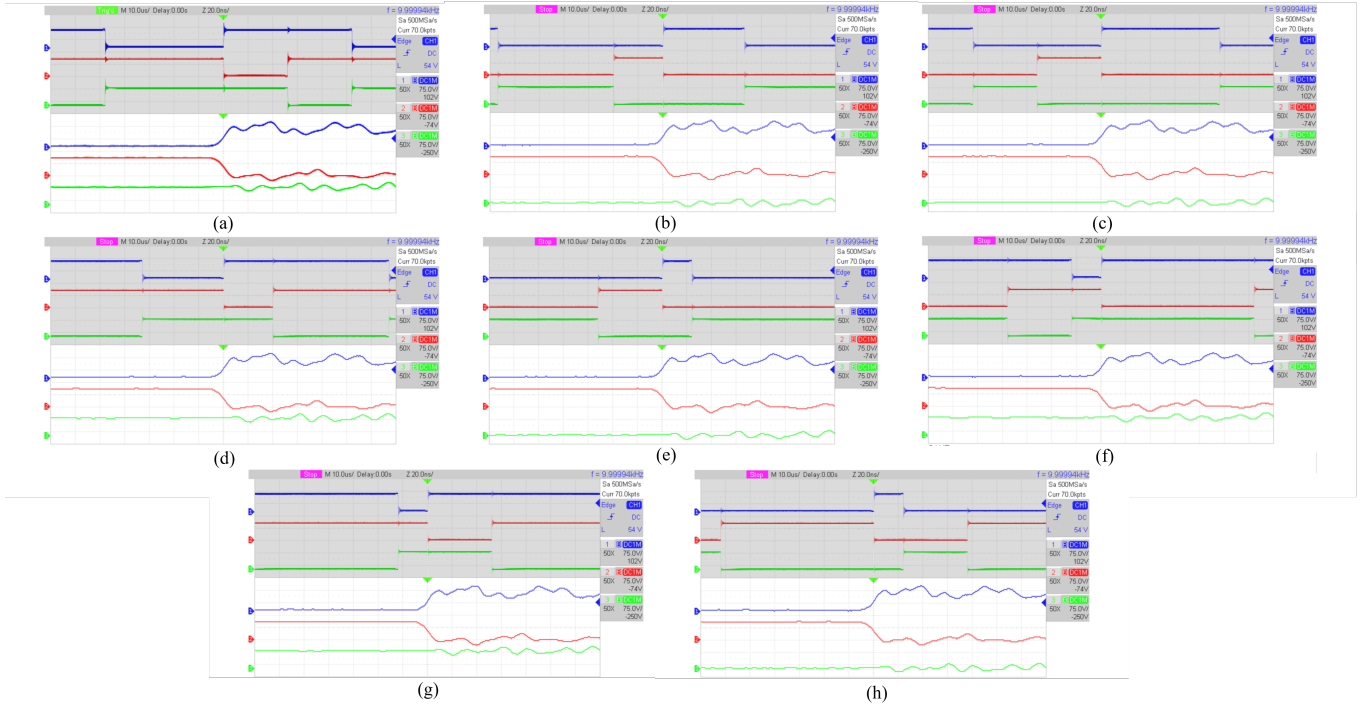
technique, and the variable carrier phase shifts that are updated for each switching period are introduced. In this way, complete dv/dt cancelation is achieved in an ideal case. The main contributions of the study are characterized by:

the MOSFETs. However, selecting the switching frequency and speed requires a holistic approach to optimize the overall system. For instance, the switching frequency is chosen based on the motor's requirements for differential current ripple and harmonic losses, whereas the switching speed is selected based on the inverter's electrical loss and thermal constraints. Consequently, utilizing

symmetric sets, such as triangular or hexagonal configurations, to enable overall cancellation. However, since the phasors depend on the harmonic number (n), a solution tailored for one harmonic may not generalize to all harmonic numbers



a close-up analysis was also performed at one operating point, focusing on two switching periods, which is shown in Fig. 9. It was observed that the proposed method effectively reduces the high-frequency common-mode voltage components while introducing a low-frequency (third-harmonic) square



PWM signals of the inverter legs, facilitated through third harmonic square wave injection and adaptive carrier phase shift, the proposed method mitigates the CMV issue. In this study, a comprehensive analysis of the modulation by the Fourier series was given, and analytical